

UĞUR ARIKAN

PRINCIPAL OPERATIONS RESEARCH SCIENTIST

Bonn, Germany

+49 171 3015770

orx.ugur.arikan@gmail.com

linkedin.com/in/uarikan

github.com/orxfun

Operations research and data science leader with 12+ years of experience turning mathematical optimization, network science, and machine learning into production decision-support systems. I lead complex planning projects from problem formulation and algorithm design through deployment and stakeholder adoption across transportation, aviation, and network operations. I bring particular expertise in Rust for high-performance and optimization software, alongside broader experience designing production systems, APIs, visualizations, and data workflows in Python, C#, TypeScript/React, and SQL.

CORE AREAS

Operations research
Optimization and modeling
Decision-support systems

Network planning
Routing and capacity
Applied data science

Software engineering
Rust and parallel computation
Open-source contribution

PROFESSIONAL EXPERIENCE

Principal Operations Research & Data Scientist | DHL Data & AI | Bonn, Germany 2018 - Present

- Lead and contribute to optimization projects across air capacity planning, network design, linehaul planning, freight consolidation, regional transport optimization, and last-mile routing, from problem formulation through deployment and operational use.
- Develop decision-support tools for air-capacity allocation and network planning, supporting strategic and tactical decisions in complex transportation systems.
- Design optimization models for network structure, facility location, routing, freight consolidation, and container loading, combining exact methods, heuristics, and scenario analysis.
- Architect and standardize reusable production engines and services for network optimization, routing, and loading, improving consistency, maintainability, and reuse across analytical applications.
- Work closely with business, network, and technical stakeholders to translate planning questions into tractable mathematical models and usable decision-support applications.
- Deliver cloud-based optimization solutions to support broader deployment, scalability, and sustainable operational use across teams and workflows.

Postdoctoral Researcher | Singapore University of Technology and Design | Singapore 2014 - 2018

- Conducted research in transportation systems, utility theory, discrete choice models, and robust optimization; supervised student projects and contributed to multidisciplinary design teams.

SOFTWARE ENGINEERING & OPEN SOURCE

- Extend this engineering work through open-source contributions in optimization, numerical methods, and data-driven planning, with attention to documentation, testing, benchmarking, and reusable abstractions. The orx-parallel repository and broader orx-* ecosystem provide public examples, with crates published via crates.io.
- Led the adoption of Rust for optimization-focused software within DHL Data & AI, establishing a production-oriented approach that enabled live Rust-based solutions for operational decision support.

EDUCATION

PhD, Industrial Engineering & Operations Research | Middle East Technical University **2009 - 2014**

Dissertation: Airline Disruption Management

MSc, Industrial Engineering & Operations Research | Middle East Technical University **2006 - 2009**

Dissertation: Two-Sided Assembly Line Balancing: Models and Heuristics

BSc, Industrial Engineering & Operations Research | Middle East Technical University, **2002 - 2006**

Ankara, Turkey

FURTHER TRAINING

Modeling and Simulation of Transportation Networks | Massachusetts Institute of **Jul 2015**

Technology

Operations Research Peripatetic Postgraduate Program | JKU Linz **Jul 2013**

RESEARCH & PUBLICATIONS

Research and peer-reviewed publications in airline disruption management, network-based recovery, traffic equilibrium, and applied optimization in logistics and transportation.

- **Flight Network-Based Approach for Integrated Airline Recovery with Cruise Speed Control**

INFORMS Transportation Science

- **Integrated Aircraft and Passenger Recovery with Cruise Time Controllability**

Annals of Operations Research

- **On the flexibility of using marginal distribution choice models in traffic equilibrium**

Transportation Research Part B: Methodological

- **Distributionally Robust Markovian Traffic Equilibrium**

INFORMS Transportation Science

- **Human-Centric Parcel Delivery with Operations Research and Machine Learning**

INFORMS Interfaces

SELECTED TALK

A practitioner perspective on applying Rust design principles to optimization software.

- **Composing Zero-Cost Abstractions in Route Optimization**

Talk on the intersection of Operations Research and Rust

SELECTED TECHNICAL CAPABILITIES

OPTIMIZATION

Mixed-integer optimization
Network optimization
Routing and facility location
Heuristics, decomposition, and local search
Scenario analysis and decision support

ENGINEERING

Rust, Python, C#, TypeScript/React, SQL
Parallel computation and concurrency
Production service design
Cloud deployment and integration
Profiling, benchmarking, and reusable library design

LANGUAGES

English (fluent) | German (intermediate) | Turkish (native)